

R&D Project

Thermal Analysis of GaN HEMT

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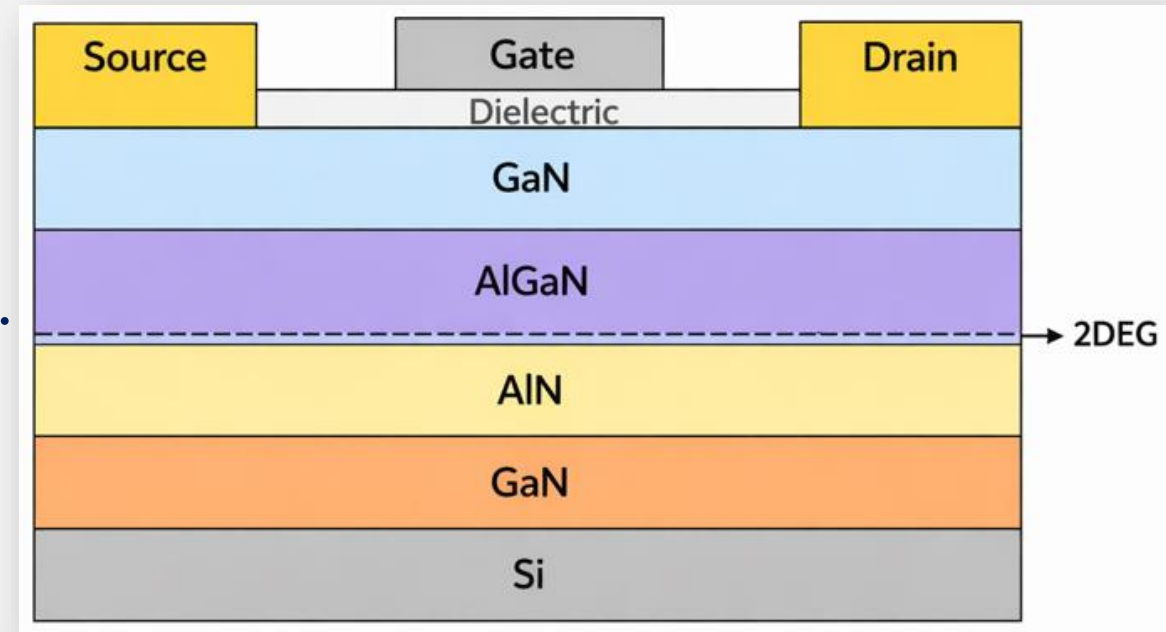
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OBJECTIVE

To analyze the thermal behavior of GaN HEMT devices under different operating conditions and substrate materials, and to evaluate heat distribution, hotspot formation, and thermal resistance in order to improve device reliability and performance.

GaN HEMT

- Gallium Nitride High Electron Mobility Transistor.
- High-power, high-frequency device.
- Based on AlGaN/GaN heterostructure.
- Used in RF, 5G, power electronics.



• Working Principle:

A GaN HEMT operates using an AlGaN/GaN heterostructure where polarization effects create a high-density 2DEG (Two-Dimensional Electron Gas) at the interface. This 2DEG acts as a high-mobility channel for electron flow between source and drain. The gate voltage controls this channel, enabling or blocking current conduction.

Heat Flow in GaN HEMT

- Heat is generated in the channel region (2DEG).
- Flows vertically through GaN → substrate → ambient.
- Substrate acts as the main heat dissipation path.
- Higher thermal conductivity → faster heat spreading.
- Poor substrates lead to heat accumulation.

Self-Heating

- Self-heating is the temperature rise in the device due to power dissipation.
- Occurs at high power density regions (mainly near gate edge).
- Caused by Joule heating and limited heat dissipation.
- Leads to reduced performance and reliability issues.

Why Self-Heating Analysis is Necessary:

- To estimate temperature rise (ΔT) in the device.
- To identify hotspot locations.
- To ensure safe and reliable operation.
- Helps in material selection and thermal optimization.

Thermal Resistance

- Thermal resistance represents the ability of a device to dissipate heat.
- Defined as the ratio of temperature rise to power dissipation.
- $R_{th} = \Delta T / P_d$
- Lower $R_{th} \rightarrow$ better heat dissipation and cooling.
- Strongly depends on substrate material and geometry.
- Critical for thermal management and device reliability.

Why ANSYS

- Provides accurate thermal and multiphysics simulation.
- Handles complex device geometries and heat flow.
- Offers clear visualization of temperature and hotspots.

ANSYS vs TCAD:

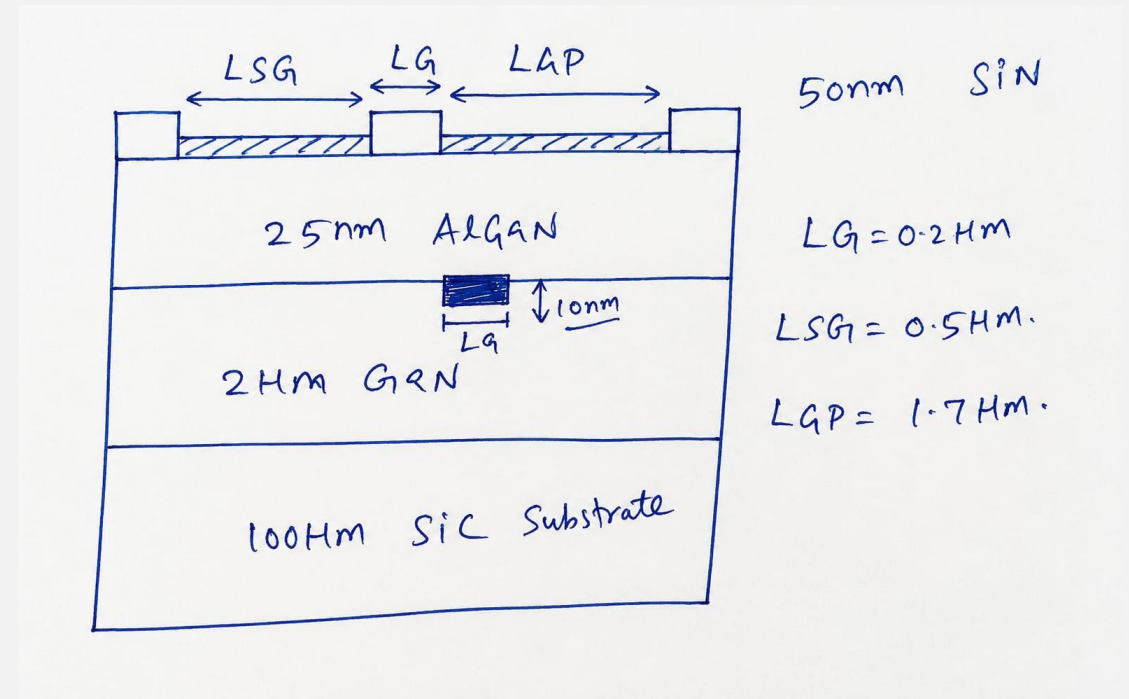
- ANSYS focuses on thermal & multiphysics, TCAD on device physics.
- ANSYS is better for heat analysis, TCAD for electrical behavior.
- ANSYS gives faster and clearer thermal results for this study.

Device Structure

- **SiN:** $0.05\text{ }\mu\text{m}$
- **AlGaN layer:** $0.025\text{ }\mu\text{m}$
- **GaN layer:** $2\text{ }\mu\text{m}$
- **SiC Substrate:** $100\text{ }\mu\text{m}$
- **LG (Gate length):** $0.2\text{ }\mu\text{m}$
- **LSG (Source–Gate spacing):** $0.5\text{ }\mu\text{m}$
- **LGD (Gate–Drain spacing):** $1.7\text{ }\mu\text{m}$

Hotspot Dimensions

- **Width:** $0.2\text{ }\mu\text{m}$
- **Depth:** $0.01\text{ }\mu\text{m}$



Effect of Hotspot Dimensions on Temperature

- Heat is generated in localized hotspot region.
- Smaller hotspot → higher temperature.
- Larger hotspot → better heat spreading.
- $\Delta T \propto P/A$
- P: Power dissipation, A: Hotspot area
- Increasing hotspot size reduces thermal resistance and temperature.

Thermal Conductivity

Thermal Conductivity (W/m.K)

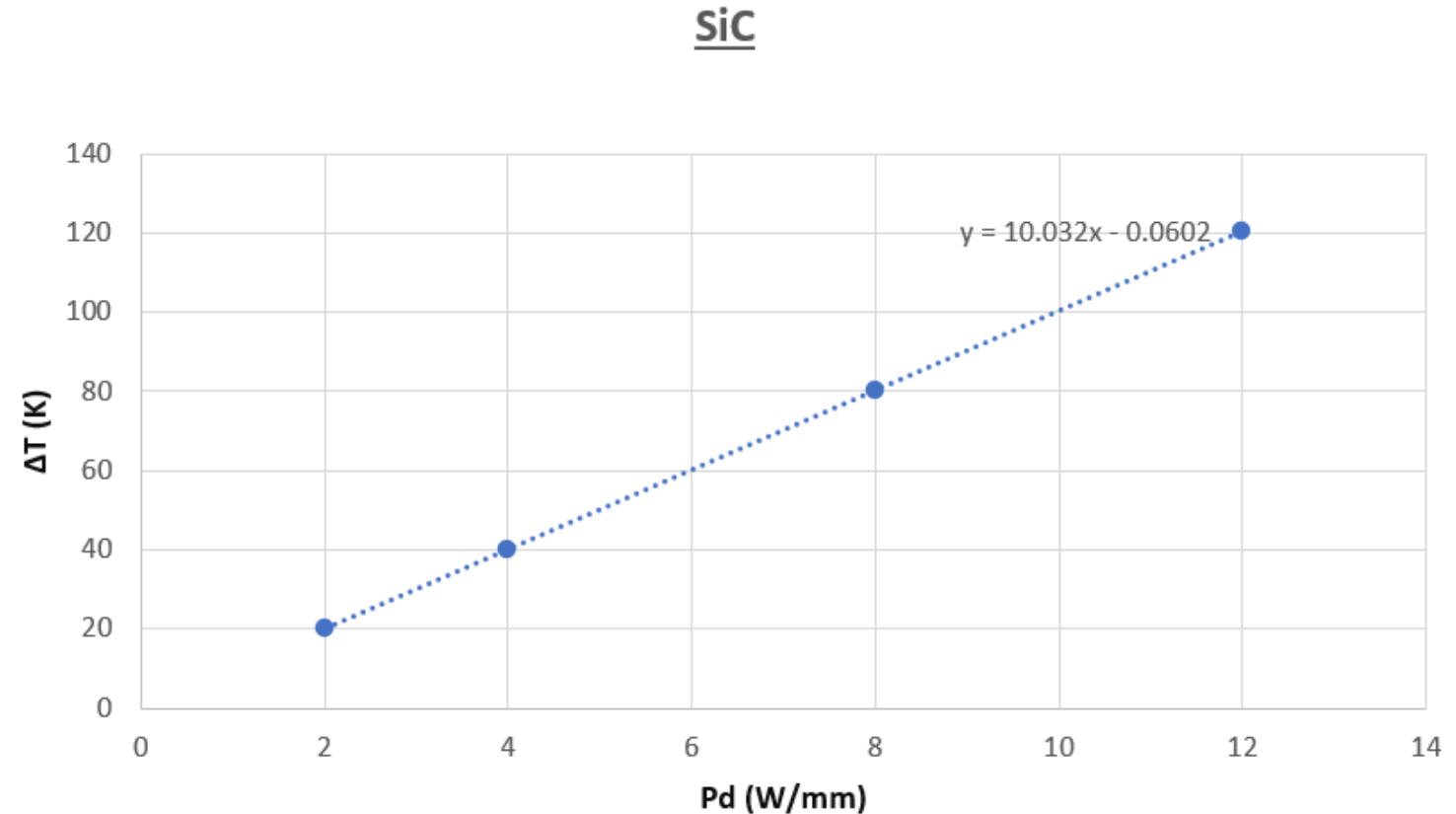
- AlGaN $\rightarrow$ 19
- GaN $\rightarrow$ 130
- Au $\rightarrow$ 317
- Diamond $\rightarrow$ 2000
 - SiC $\rightarrow$ 400
 - Si $\rightarrow$ 148
- Sapphire $\rightarrow$ 35
 - SiN $\rightarrow$ 33

RESULTS FOR DIFFERENT SUBSTRATES

- FOR SiC:

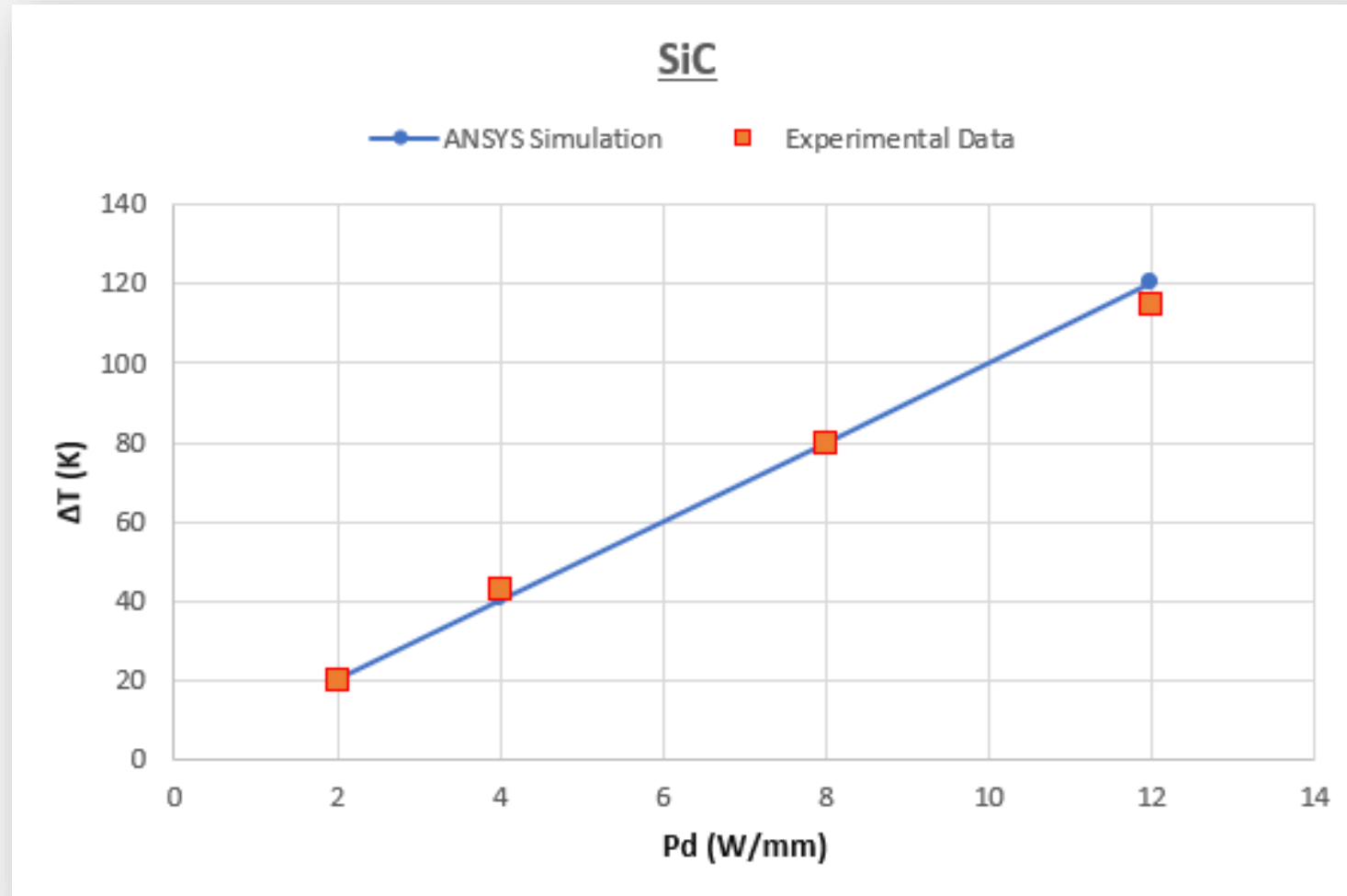
SiC				
Heat Generation (W/ μm^3)	Max Temperature (K)	Min Temperature (K)	Pd (W/mm)	ΔT (K)
1.00E+11	320.02	300	2	20.02
2.00E+11	340.04	300	4	40.04
4.00E+11	380.21	300	8	80.21
6.00E+11	420.32	300	12	120.32

SiC Rth : 10.03 K·mm/W



COMPARISON

- Comparison of simulation results with reference data (SiC)

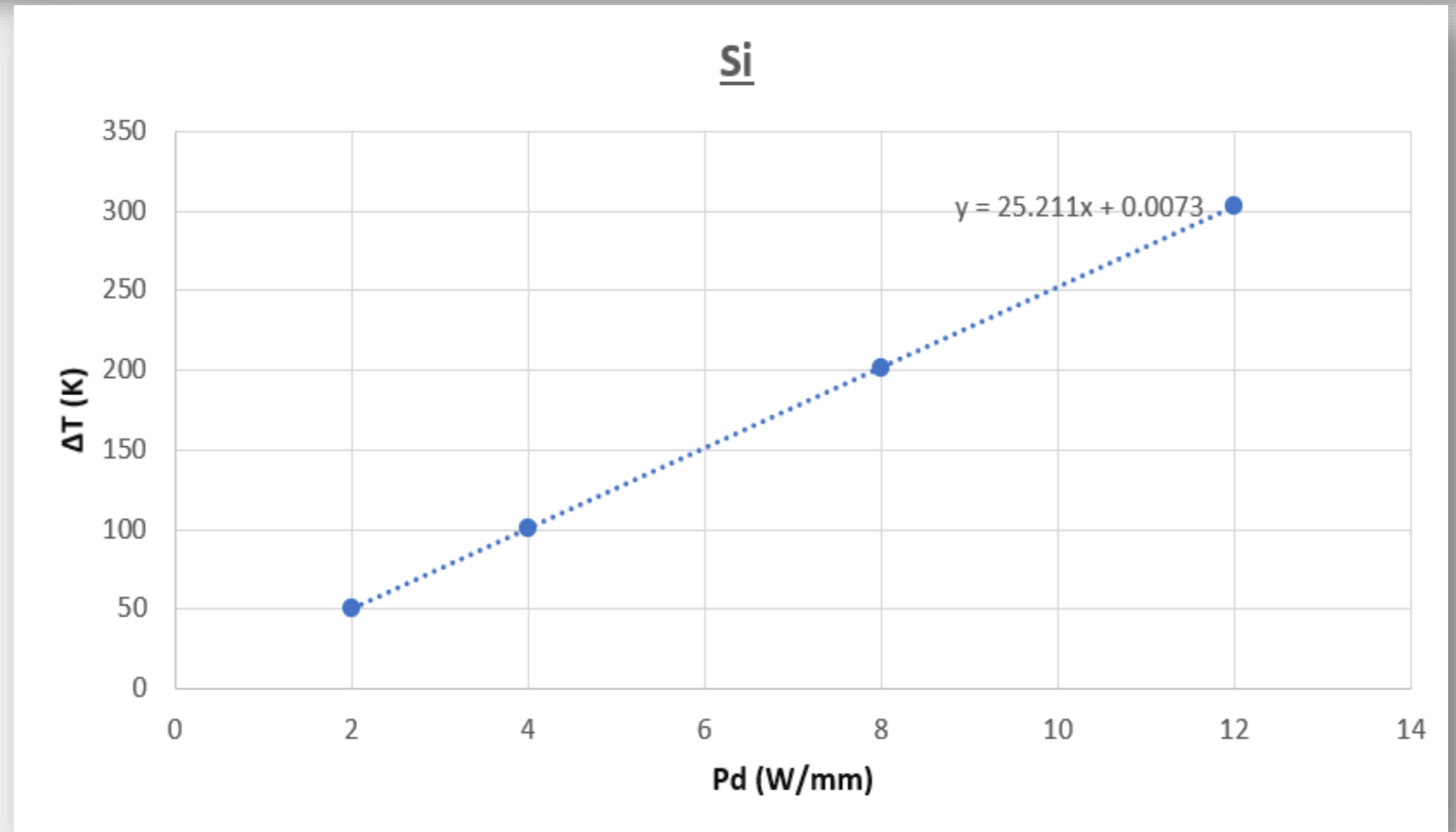


Journal paper reference : K. S. Powar, V. K. Tadepalli, V. Vijay Painter, R. Sommet, A. Chakravorty, P. V. Raja, Maximum, effective, and average thermal resistance for GaN-based HEMTs on SiC, Si, and sapphire substrates, Solid-State Electron. 227 (2025) 109121

- **FOR Si:**

Si					
Heat Generation (W/ μm^3)	Max Temperature (K)	Min Temperature (K)	Pd (W/mm)	ΔT (K)	
1.00E+11	350.43	300	2	50.43	
2.00E+11	400.85	300	4	100.85	
4.00E+11	501.7	300	8	201.7	
6.00E+11	602.54	300	12	302.54	

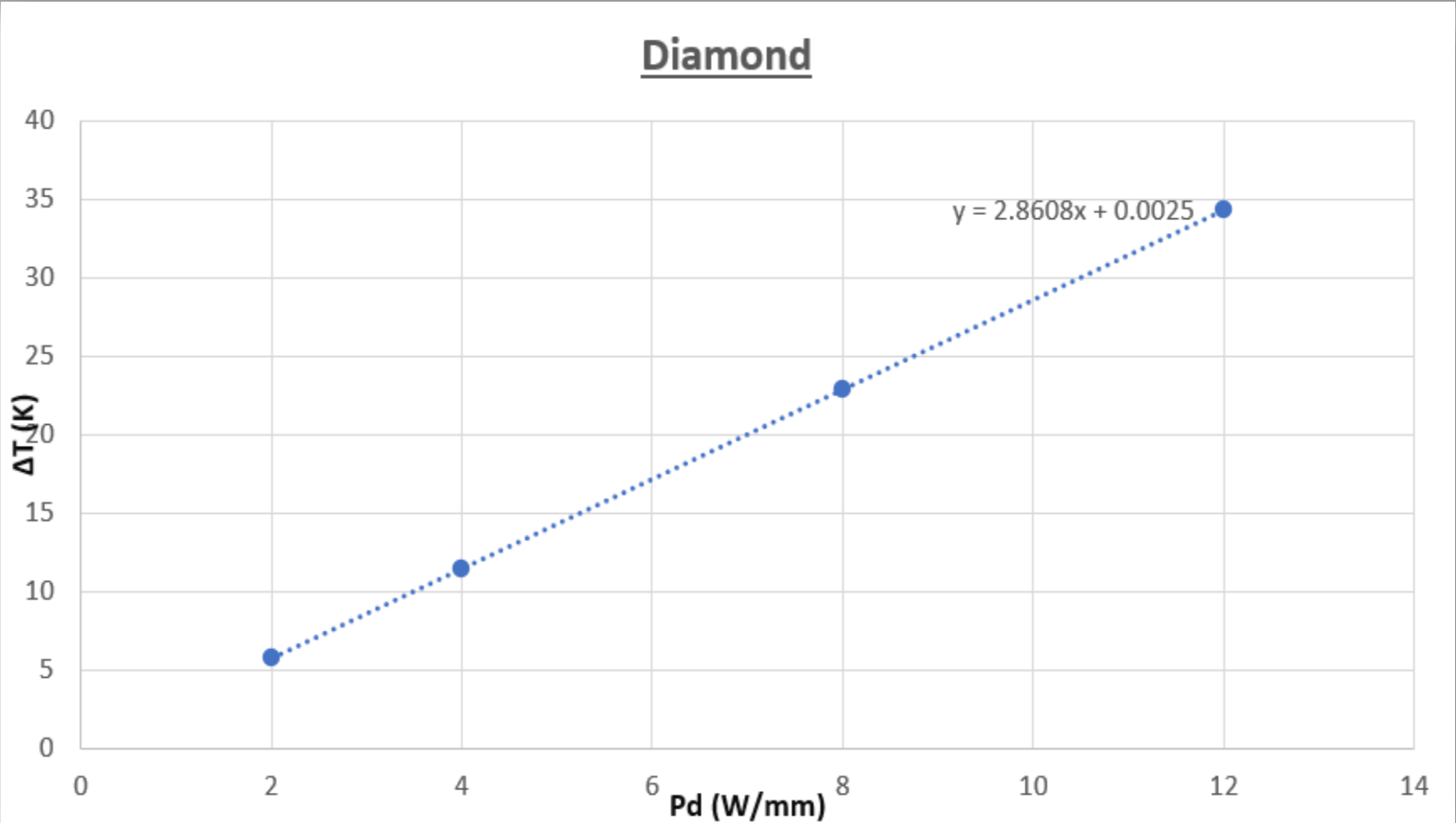
Si Rth = 25.21 K·mm/W



• FOR DIAMOND:

Diamond				
Heat Generation (W/μm³)	Max Temperature (K)	Min Temperature (K)	Pd (W/mm)	ΔT(K)
1.00E+11	305.72	300	2	5.72
2.00E+11	311.45	300	4	11.45
4.00E+11	322.89	300	8	22.89
6.00E+11	334.33	300	12	34.33

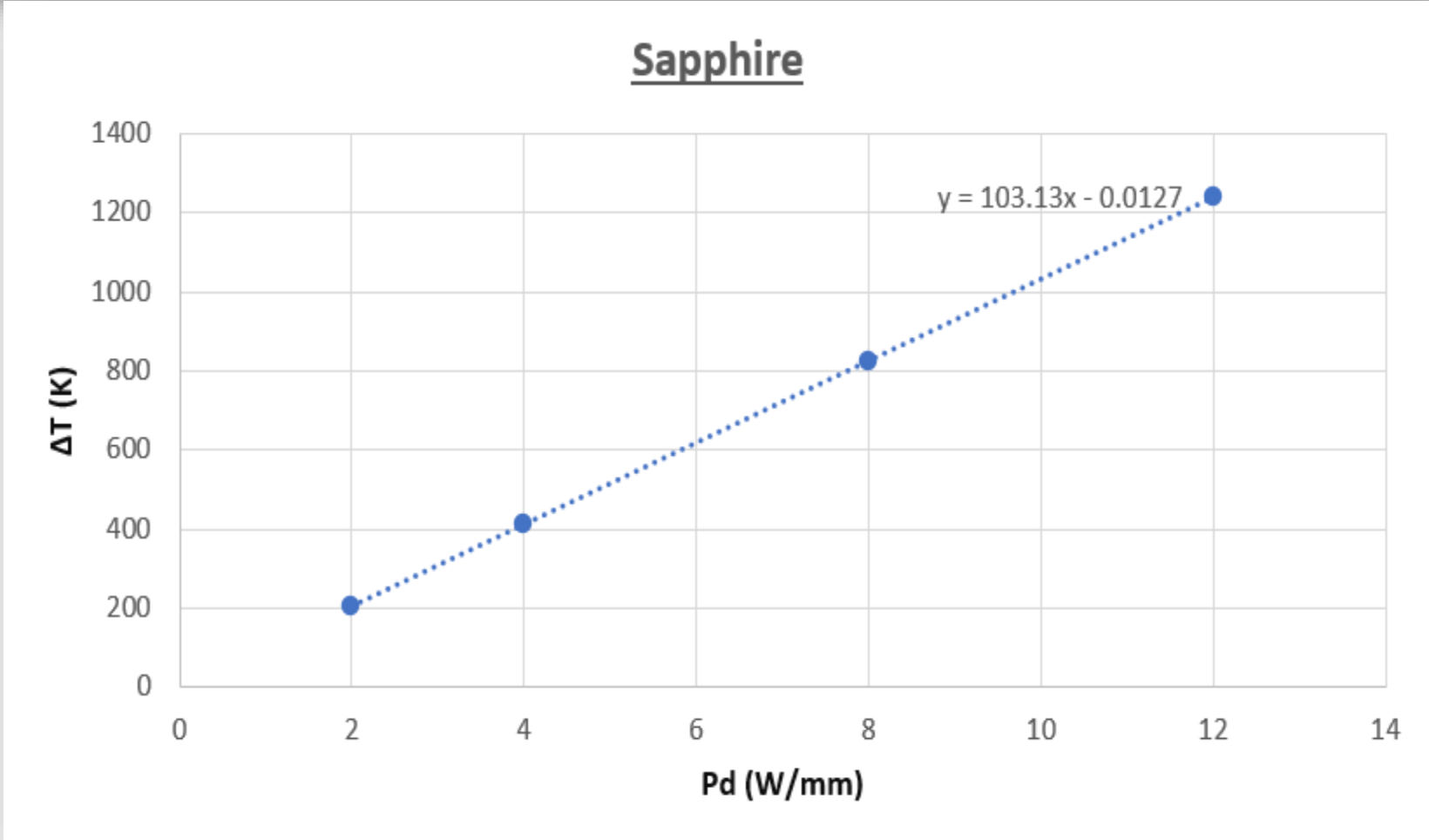
Diamond Rth = 2.86 K·mm/W



• FOR SAPPHIRE:

Sapphire				
Heat Generation (W/μm³)	Max Temperature (K)	Min Temperature (K)	Pd (W/mm)	ΔT(K)
1.00E+11	506.24	300	2	206.24
2.00E+11	712.49	300	4	412.49
4.00E+11	1125	300	8	825
6.00E+11	1537.5	300	12	1237.5

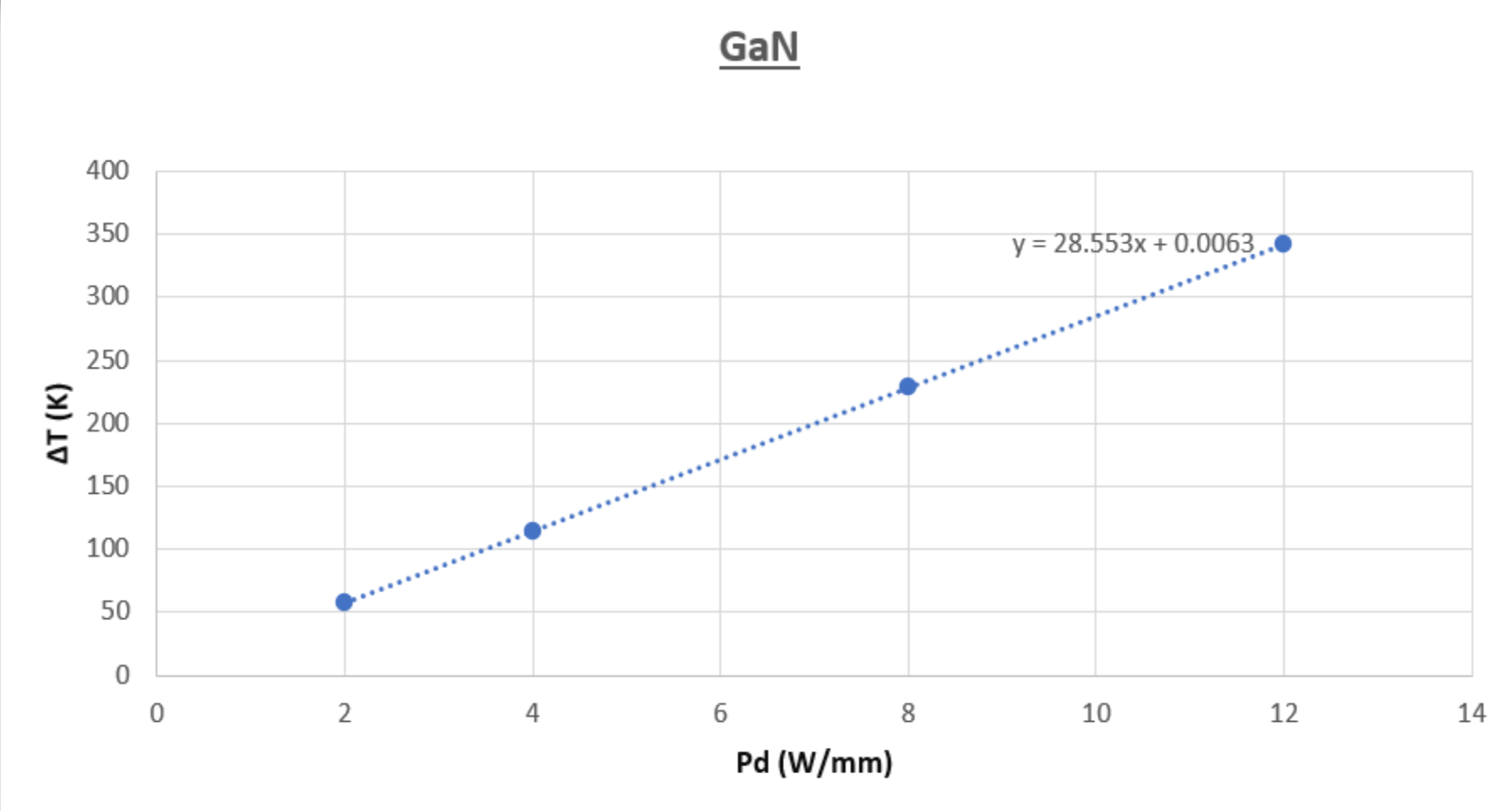
Sapphire Rth = 103.13 K·mm/W



• FOR GaN:

GaN				
Heat Generation (W/μm³)	Max Temperature (K)	Min Temperature (K)	Pd (W/mm)	ΔT(K)
1.00E+11	357.11	300	2	57.11
2.00E+11	414.22	300	4	114.22
4.00E+11	528.43	300	8	228.43
6.00E+11	642.64	300	12	342.64

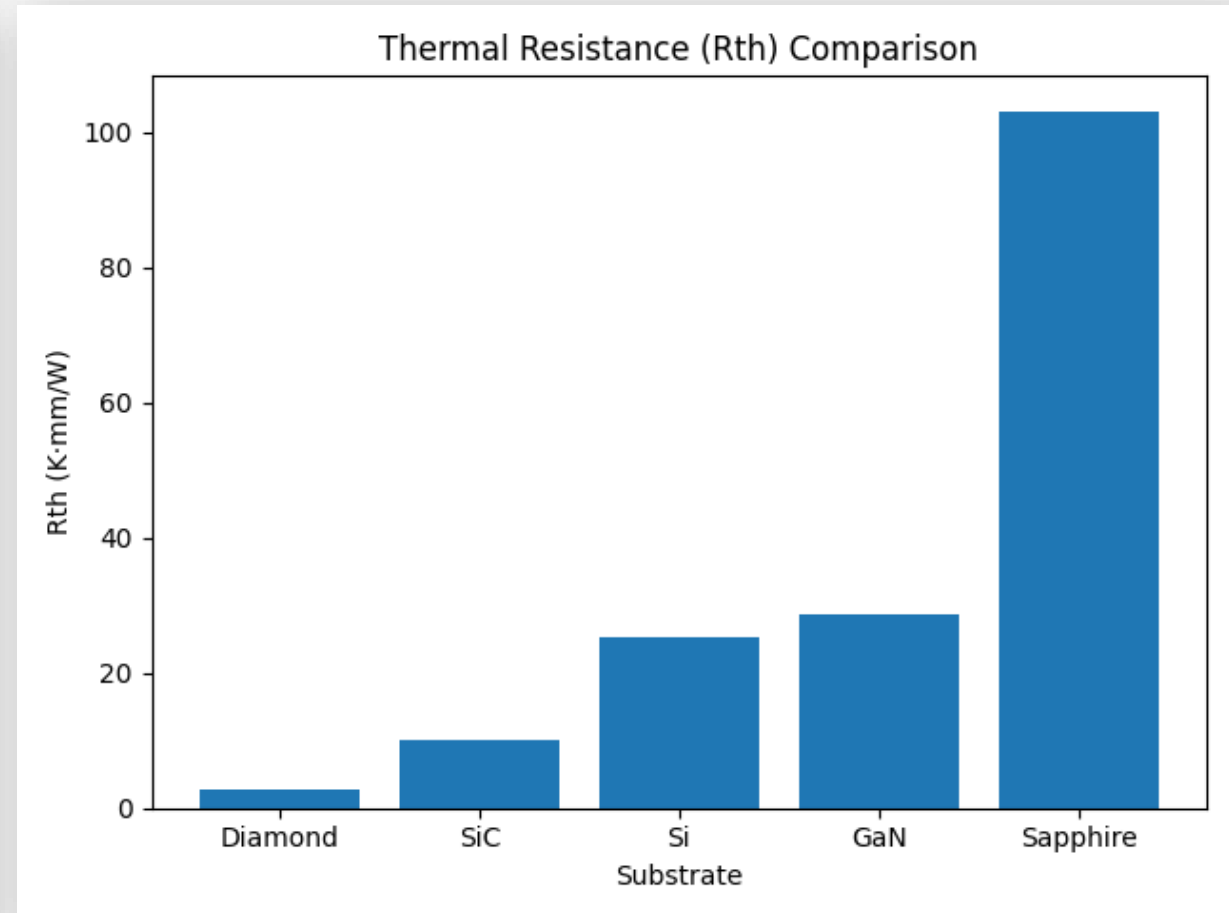
GaN Rth = 28.55 K·mm/W



Thermal Resistance (Rth) Comparison

Rth Values for 5 Substrates

Substrate	Rth (K·mm/W)
Diamond	2.86
SiC	10.03
Si	25.21
GaN	28.55
Sapphire	103.13



Lower thermal resistance results in better heat dissipation and reduced device temperature.

OBSERVATION

- **Diamond** → Best performance

Very low R_{th} due to extremely high thermal conductivity (2000 W/mK) → excellent heat dissipation.

- **SiC** → Good performance

Low R_{th} with high thermal conductivity (400 W/mK) → efficient cooling.

- **Si & GaN** → Moderate performance

Medium R_{th} due to moderate thermal conductivity (130–150 W/mK) → limited heat flow.

- **Sapphire** → Worst performance

Very high due to low thermal conductivity (35 W/mK) → poor heat dissipation.

Temperature Comparison for Different Power Levels

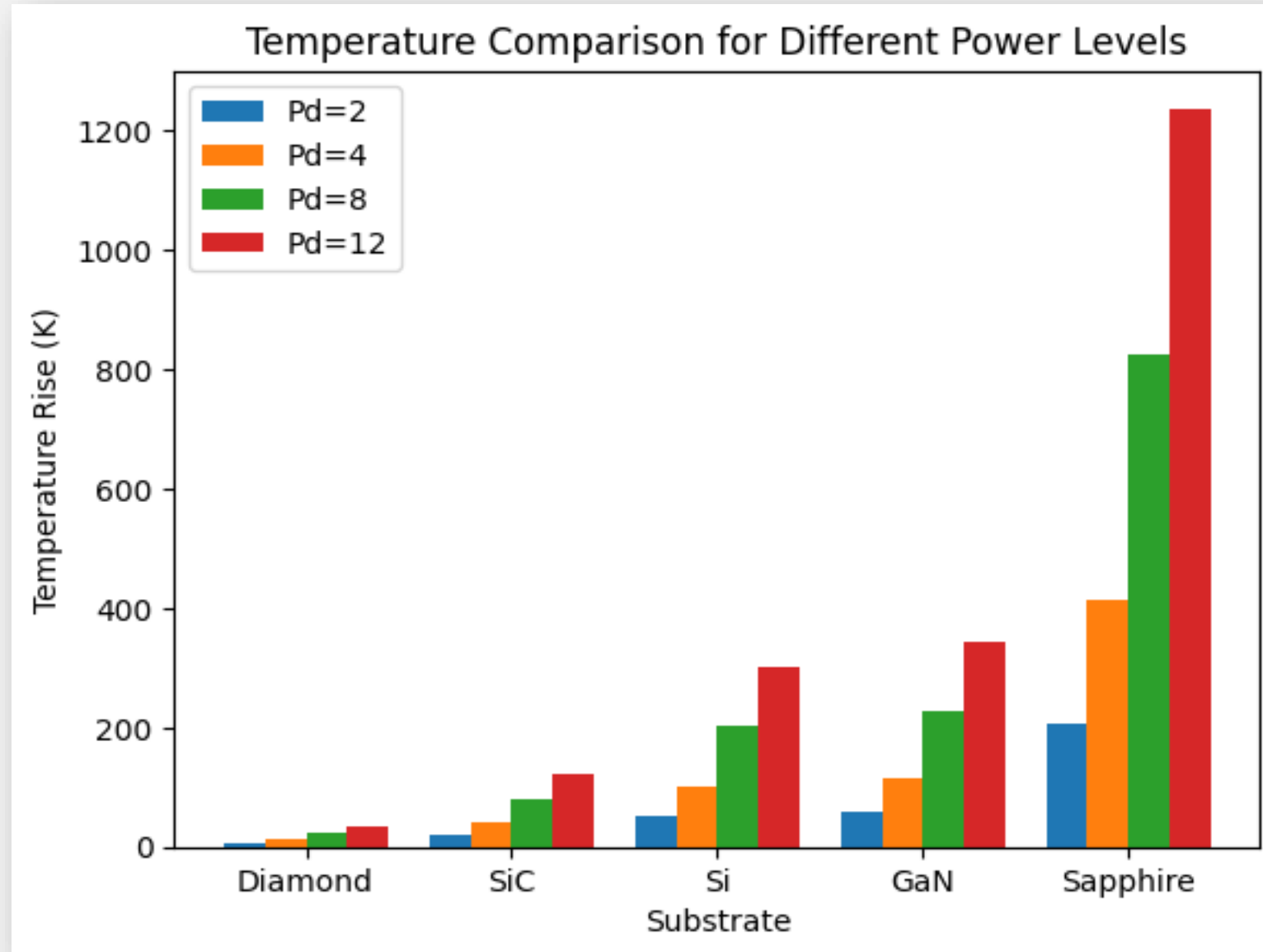


Figure 1: Thermal conductivity structure in different temperature ranges

Temperature vs Power Relation

- Temperature rise increases linearly with power density.
- $\Delta T \propto P_d$
- Slope of graph gives thermal resistance (R_{th}).
- Verified through simulation and Excel data.
- Confirms model accuracy.

CONCLUSION

- Thermal analysis of GaN HEMT was successfully performed using ANSYS.
- Thermal resistance varies strongly with substrate material .
- Diamond → best, SiC → good, Sapphire → worst performance .
- Simulation results are validated and consistent with theory .
- Efficient thermal management is critical for improving GaN HEMT performance and reliability.